

DOTHAN, AL, USA

WMO: 722268

Lat: 31.317N

Lon: 85.450W

Elev: 114

StdP: 99.96

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	-0.8	-11.4	1.4	2.1	-8.8	1.8	3.7	10.9	12.5	9.8	10.0	3.4	320	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	35.9	24.2	34.4	24.0	33.4	23.9	26.6	31.8	26.0	31.1	25.5	30.4	3.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	28.4	24.8	20.1	28.0	24.1	19.3	27.5	83.4	32.0	80.8	31.2	78.6	30.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	8.0	7.0	DB	-6.2	37.1	2.2	1.6	-7.8	38.3	-9.1	39.2	-10.3	40.2	-12.0	41.3	(4)
(5)			WB	-7.5	27.2	2.1	0.8	-9.1	27.8	-10.3	28.3	-11.5	28.7	-13.0	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	10.1	12.4	15.6	19.3	24.0	27.0	27.9	27.5	25.3	20.2	14.7	11.6
	DBStd	7.43	5.56	5.04	4.52	3.55	2.95	2.16	1.65	1.77	2.58	4.03	4.54	5.22
	HDD10.0	162	69	30	9	0	0	0	0	0	0	1	10	43
	HDD18.3	944	259	174	106	31	3	0	0	0	1	28	127	216
	CDD10.0	3692	72	96	183	281	433	511	554	542	458	317	151	93
	CDD18.3	1432	4	7	21	61	178	261	296	284	209	87	18	7
	CDH23.3	13139	10	34	158	509	1642	2552	2943	2728	1769	675	97	22
CDH26.7	5521	0	2	18	121	668	1197	1349	1226	741	191	9	1	
Wind	WSAvg	3.2	3.7	3.7	3.8	3.6	3.2	2.9	2.6	2.5	2.8	3.0	3.1	3.4
Precipitation	PrecAvg	1353	128	111	140	100	72	132	153	112	101	73	107	97
	PrecMax	1644	326	216	292	243	175	266	475	251	374	164	281	216
	PrecMin	1028	26	31	5	22	1	45	63	32	30	1	38	36
	PrecStd	202	71	52	78	58	46	72	91	61	87	49	60	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.3	28.6	31.2	35.1	37.6	37.2	37.0	35.5	32.2	27.8	25.7
		MCWB	18.8	18.8	18.6	21.2	22.2	24.6	24.6	25.1	23.1	22.7	20.3	19.8
	2%	DB	22.6	24.1	27.0	29.4	33.0	35.8	35.1	34.9	33.7	30.3	26.1	23.4
		MCWB	17.8	17.9	18.1	20.0	21.8	24.0	24.7	25.1	23.0	21.6	19.4	19.0
	5%	DB	20.9	22.4	25.2	27.9	31.9	34.0	33.9	33.6	32.3	28.8	24.1	21.7
		MCWB	17.4	17.1	17.5	19.3	21.4	23.4	24.6	24.8	23.3	20.9	18.3	18.5
	10%	DB	18.8	20.4	23.4	26.6	30.4	32.5	32.7	32.4	31.0	27.3	22.5	19.9
		MCWB	16.0	16.2	16.4	18.5	20.9	23.2	24.5	24.7	23.1	20.0	17.9	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.6	21.4	23.3	25.0	26.9	27.4	27.5	25.9	25.2	22.9	21.8
		MCDB	22.0	23.2	24.2	27.6	30.2	33.0	32.2	32.7	30.2	28.5	24.9	23.6
	2%	WB	19.3	19.7	20.3	22.2	23.9	26.0	26.4	26.7	25.2	23.9	21.6	20.4
		MCDB	21.5	22.8	23.4	26.4	28.6	32.0	31.9	31.6	29.5	27.1	23.5	22.0
	5%	WB	18.0	18.6	19.3	21.2	23.1	25.3	25.8	26.1	24.7	22.9	20.3	19.2
		MCDB	20.0	21.1	22.8	25.2	27.8	30.6	31.0	30.8	29.1	26.2	22.9	21.0
	10%	WB	16.6	17.3	18.3	20.3	22.5	24.6	25.3	25.5	24.2	22.0	18.9	17.8
		MCDB	18.6	19.5	21.8	24.2	27.2	29.6	30.4	30.2	28.7	25.3	21.7	19.6
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.2	12.0	12.1	11.7	10.2	9.9	9.8	10.5	12.0	12.2	10.6
		MCDBR	11.3	12.1	13.3	13.2	12.7	12.3	11.2	11.1	12.1	12.6	12.4	10.9
		MCWBR	7.5	7.0	6.2	5.3	3.8	3.4	3.3	3.6	4.1	5.2	6.7	7.2
	5% WB	MCDBR	10.1	10.1	10.6	10.4	10.4	9.9	9.9	9.8	9.7	9.5	9.7	9.7
		MCWBR	7.7	7.0	6.4	5.5	4.0	3.7	3.5	3.6	3.9	4.9	6.4	7.6
Clear-Sky Solar Irradiance	taub	0.323	0.332	0.354	0.381	0.431	0.470	0.508	0.498	0.428	0.370	0.346	0.329	
	taud	2.520	2.494	2.436	2.367	2.256	2.212	2.116	2.140	2.316	2.452	2.466	2.521	
	Ebn at Noon	906	930	928	911	862	825	791	795	842	875	870	880	
	Edh at Noon	86	97	110	123	138	144	158	152	123	100	91	83	
All-Sky Solar Radiation	RadAvg	3.00	3.58	4.78	5.86	6.41	6.08	5.95	5.53	4.99	4.33	3.37	2.60	
	RadStd	0.18	0.44	0.39	0.48	0.46	0.47	0.37	0.37	0.38	0.47	0.28	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-93	+166	+109

Nomenclature: See separate page